Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2024/2025

COURSE NAME	:	HUMAN-COMPUTER INTERACTION
COURSE CODE	:	BIM 30503
PROGRAMME CODE	:	BIM/BIP/BIS/BIW
EXAMINATION DATE	:	FEBRUARY 2025
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

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THIS QUESTION PAPER CONSISTS OF SEVEN (7) PAGES

PART A

Instruction: Answer **ALL** questions in the answer booklet.

Q1 Who are the Stakeholders in the Higher Education System in Malaysia?

(i) Lecturers and students at UTHM.
(ii) Administration staff at UTHM.
(iii) Parents of students who study at UTHM.
(iv) Industries in Malaysia.

(a) i.
(b) i, ii, and iii.
(c) i, ii, and iv.
(d) All the above.

(1 mark)

Q2 What role does empathy play in user-centered design (UCD) ?

(a) Analyzing market trends to predict user behavior.
(b) Consulting experts and using technical data to develop solutions.
(c) Observing, engaging, and empathizing with users to understand their experiences and motivations.
(d) Designing solutions based solely on personal assumptions about user needs.

(1 mark)

Q3 Which of the following does **NOT** describe "user need" in UCD?

(a) Focusing on aesthetic design.
(b) Focusing on user's goals.
(c) Focusing on user experience.
(d) Focusing on usability.

(1 mark)

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Q4 What is the primary focus of evaluation in the design process of a system?

(a) To analyse the technical specifications of the system.

(b) To improve the marketing strategies of the product.

(c) To enhance the usability of the system and the users' experience.

(d) To assess the budget and resource allocation for the project.

(1 mark)

Q5 An experiment is one of the data gathering methods used in usability testing. Which of the following actions can be conducted during the experiment on the MySejahtera application?

(i) Testing with real user health data or personal information.

(ii) Observe the number of mistakes and kinds of errors made by the users.

(iii) Experimentally removing control from users by turning off the back button.

(iv) Test the accessibility of features for users with different needs to ensure inclusivity.

(a) i & iii.

(b) ii & iv.

(c) ii, iii & iv.

(d) All the above.

(1 mark)

Q6 Which evaluation method involves a process where experts use rules of thumb to measure the usability of user interfaces in independent walkthroughs and report issues?

(a) Analysis.

(b) Inspections.

(c) Cognitive walkthrough.

(d) Heuristics evaluation.

(1 mark)

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Q7 What is the priority consideration when choosing the evaluation method for the MySejahtera application?

(a) The choice depends on what goal is to be achieved by the organisation.

(b) The choice depends on which user will be involved in the evaluation.

(c) The choice depends on what data is required by the organisation.

(d) The choice depends on which location the evaluation will be conducted.

(1 mark)

Q8 Which of the following is **NOT** considered to be an essential requirement for effective user support in Human-Computer Interaction (HCI)?

(a) Availability.

(b) Robustness.

(c) Unobtrusiveness.

(d) Personalization.

(1 mark)

Q9 Determine which statement accurately describes the strengths and weaknesses of live chat support.

(a) Live chat provides immediate responses and real-time interaction but may have limited availability and lead to less detailed communication.

(b) Live chat support is always available 24/7, ensuring immediate responses and thorough communication at all times.

(c) Live chat typically results in longer wait times, while also providing lower user satisfaction due to its lack of real-time interaction.

(d) Live chat interactions prioritize thoroughness over speed, leading to immediate responses and higher user engagement.

(1 mark)

Q10 _____ is **NOT** a method for displaying help in a software application.

(a) New window

(b) Whole screen or split screen

(c) Pop-up box

(d) Audio instructions

(1 mark)

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Q11 What is a crucial characteristic of UCD ?

(a) It emphasises aesthetic design over functionality.

(b) It is a linear process with fixed stages.

(c) It involves an iterative approach that prioritises understanding users and their context.

(d) It is only applicable in the final stages of product development.

(2 marks)

Q12 Which of the following statements are **TRUE** for the design thinking process?

(i) Understanding the human needs involved is a central focus.

(ii) Engaging users is optional and can be substituted with expert opinions.

(iii) Adopting a hands-on approach in prototyping and testing.

(iv) Rigid adherence to a fixed sequence of phases is essential for success.

(a) ii & iii.

(b) i & iii.

(c) ii, iii & iv.

(d) All the above.

(2 marks)

Q13 The MySejahtera application was designed to help manage public health, particularly in response to the COVID-19 pandemic in Malaysia. Before releasing the MySejahtera application to users, which evaluation type would be most suitable?

(a) Heuristic evaluation.

(b) Natural setting involving users.

(c) Any setting not involving users.

(d) Controlled setting involving users.

(2 marks)

Q14 A software company is designing a new task management tool for freelancers. The team conducted interviews and observations with potential users to develop personas. One key persona is "Freelancer Fiona", a graphic designer who juggles multiple projects with tight deadlines. Fiona values simplicity in a tool, prefers mobile access, and often works from various locations like coffee shops. She finds it difficult to manage overlapping project deadlines and seeks a solution that integrates easily with her calendar and email.

Based on the case study, which of the following features would be most aligned with the needs of the persona "Freelancer Fiona"?

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(a) A feature that provides in-depth analytics on project performance.

(b) A desktop-only interface with multiple complex customization options.

(c) A mobile-friendly task management interface that integrates with calendars and email.

(d) A tool focused on long-term strategic planning with little emphasis on day-to-day tasks.

(2 marks)

Q15 An e-commerce company, "ShopEase" has been receiving feedback that customers are finding it difficult to track their orders after making a purchase. The company currently provides a FAQ section on their website, but it lacks specific details on order tracking. After reviewing the issue, the user support team is considering adding context-sensitive help to their order tracking system and implementing live chat support to address real-time customer inquiries. Additionally, they plan to provide a dedicated knowledge base with tutorials and step-by-step instructions for common support issues.

Based on the case study, which of the following would be the most effective user support improvement for addressing customers' difficulties with order tracking at ShopEase?

(a) Expanding the FAQ section to cover more topics.

(b) Adding context-sensitive help within the order tracking page, along with live chat support for real-time assistance.

(c) Sending automated emails to customers with instructions after every purchase.

(d) Requiring customers to call a support line for all order-related inquiries.

(2 marks)

PART B

Q16 Suppose a tech company, SecureHome, aimed to develop a smart home security system. Understanding the importance of user-centered design (UCD), they decided to involve potential users throughout the design process.

(a) Explain stages in UCD with an appropriate diagram.

(8 marks)

(b) Considering the importance of UCD, suggest **FOUR (4)** actions the SecureHome company would take to realise the involvement of potential users throughout the design process. Provide an appropriate example for each action.

(12 marks)

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Q17 Evaluation is important to the design process, focusing on the system's usability and users' experience when interacting with the system.

(a) Compare the user evaluations conducted in controlled settings versus natural settings.

(8 marks)

(b) Analyse **FOUR (4)** usability issues that could be identified during testing of an online shopping website and how each issue impacts the user experience.

(12 marks)

Q18 Designing a user support interface that optimizes the presentation of information for different user types (beginners and advanced users) requires creating a flexible, intuitive system that can adapt to users' varying needs.

(a) Differentiate **FOUR (4)** approaches of user support that allow users both beginners and advanced,

(i) to perform search functionality within the web application

(4 marks)

(ii) to give feedback and support options.

(4 marks)

(b) Draw **ONE (1)** interface consisting of **FOUR (4)** user supports for hiking mobile application specially tailored for beginners to ensure they have a safe, enjoyable, and informative experience.

(12 marks)

- END OF QUESTIONS -

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